FINITE-SAMPLE CERTIFICATES FOR ADAPTIVE STOPPING IN COUNT-EXCHANGEABLE BINARY PASS-COUNT REPLAY

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ABSTRACT

We study a count-exchangeable binary pass-count replay model for repeated model rollouts, where a task's N binary outcomes are summarized by its pass count. For the *true* count law $H_\star$, an oracle posterior score $c_{H_\star}(x,k)$ is the exact conditional replay-flip probability; the corresponding oracle stopping theorem follows by the tower property without a union bound over stopping times. The implemented BAYES-H rule instead freezes $\widehat{H}_{\mathrm{FIT}}$ and thresholds the fitted stopping score $s_{\widehat{H}_{\mathrm{FIT}}}(x,k)$. We do *not* identify that plug-in score with the oracle conditional probability. Its operative high-probability statement is Theorem 2: under independent i.i.d. FIT/CAL task sampling, a finite FIT-frozen family is CAL-screened using empirical-Bernstein UCBs with a Bonferroni allocation, yielding a marginal replay guarantee for a selected frozen rule; TEST is descriptive.

On the supplied OpenMathReasoning count-replay artifact (11,607 problems, $N = 32$; FIT/CAL/TEST $4000/4000/3607$), the CAL-selected fitted-score rule has a descriptive TEST replay rollout-count reduction of 80.9% at realized replay flip 0.0249 for $\alpha = 0.05$ and 73.5% at $\alpha = 0.02$; the corresponding FIXED-HOEF/FIXED-EB rows are reported alongside it. An ESC-style window heuristic has a 0.0577 replay flip in this TEST readout. The supplied synthetic drift and all margin-exploration results are explicitly synthetic, exploratory, and nonconfirmatory: at E3, $\alpha = 0.05$, and $\delta = 0.15$, BAYES-H is invalid (5.2% flip) while FIXED-EB remains valid (4.7%). The OpenR1 carrier is accounted for as 9,374 raw unique problems, of which 8,853 have exactly two deduplicated rollouts and are split $3000/3000/2853$ (FIT/CAL/TEST). Every reported flip is disagreement between binary pass-count replay decisions, and every saving is $1 - \bar{k}/N$; the artifacts do not measure a delivered-answer majority, gold correctness, generated tokens, latency, cancellation, or ordered online deployment.

1 INTRODUCTION

Self-consistency (Wang et al., 2023) replaces one sampled answer by a majority vote over N samples. Adaptive variants reduce or reallocate sampling: ASC (Aggarwal et al., 2023) adapts a per-problem count with a lightweight stopping criterion, ESC (Li et al., 2024) proposes early stopping for self-consistency, and MARS (Chen et al., 2026) stops parallel traces using a conservative bound on future vote movement. For a count-only replay analysis, two deployment factors remain outside its evidence: (i) within-task rollouts may be correlated; and (ii) chronological generation order may drift, breaking an exchangeability-based certificate.

We make two model-scoped observations. First, in a count-only carrier the observable and modeled task-level variation is *across-task* heterogeneity: per-task pass rates are strongly bimodal (Section 3), and the replay flip probability is a functional of the mixture of per-task counts. Because the artifact retains counts rather than ordered outcomes, it cannot identify whether within-task correlation is the dominant deviation from i.i.d., nor establish a pooled-correlation diagnostic. Second, once the target is the mixture, the correct *oracle* object is a per-prefix conditional probability under the true law $H_\star$: the posterior probability, given the observed prefix, that the full-N majority disagrees. This object is exactly computable (hypergeometric posterior), its fixed-budget expectation $\mathbb{E}_H[f_K(k)]$ is

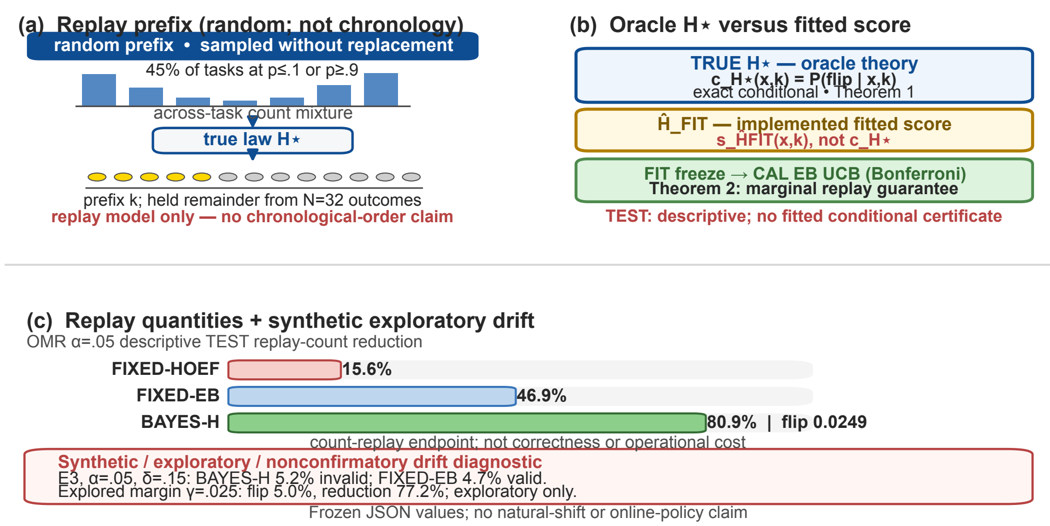

Figure 1: **Code-native overview; frozen-JSON labels.** (a) A random replay prefix is *not* chronology. (b, top) only the true count law $H_\star$ gives the exact conditional identity for $c_{H_\star}$ and the oracle tower theorem. (b, middle) BAYES-H uses the distinct fitted score $s_{\widehat{H}_{\rm FIT}}$; its operational claim is the FIT-frozen, Bonferroni CAL-EB UCB screen of Theorem 2, not an exact per-problem conditional certificate. (c) The shown replay quantities are parsed from hash-pinned frozen JSON; the drift/margin note is synthetic, exploratory, and nonconfirmatory. The figure supplies no ordered-online, correctness, or operational-cost measurement.

monotone in the prefix length (Lemma 1), and — because the tower property collapses the stopping-time dependence — it certifies *any* adaptive rule at level α with no union bound (Theorem 1). The implemented plug-in score formed with $\widehat{H}_{\rm FIT}$ is distinct; its usable guarantee is the FIT-frozen CAL-screened marginal result of Theorem 2.

Contributions.

1. **Framework.** A *binary pass-count replay* model, not an ordered online or answer-aggregation model, in which both within-task correlation and across-task heterogeneity are folded into the true mixture $H_\star$ over per-task pass counts; the target is replay flip probability (Section 3).
2. **Oracle theorem and fitted implementation.** For true $H_\star$, $c_{H_\star}(x,k)$ is an exact conditional replay-flip probability and Theorem 1 is an oracle result. BAYES-H instead thresholds $s_{\widehat{H}_{\rm FIT}}(x,k)$ after fitting $\widehat{H}_{\rm FIT}$; its operative guarantee is Theorem 2's CAL-screened marginal replay bound, not the oracle conditional identity (Section 4).
3. **Legitimate selection.** Under the stated i.i.d.-task and FIT/CAL-independence assumptions, a FIT-frozen, pre-declared rule family (budget $\times$ level for fixed-k; level for BAYES-H) is selected on CAL with simultaneous empirical-Bernstein bounds (Maurer & Pontil, 2009) plus Bonferroni. The fixed benchmark TEST split is read once and reported only descriptively (Section 5).
4. **Empirical replay readouts.** On 11,607 OpenMathReasoning problems, the one descriptive TEST readout reports 80.9% replay rollout-count reduction at replay flip 0.0249 ($\alpha = 0.05$), versus 15.6% for FIXED-HOEF; ESC-style window stopping has 0.0577 replay flip. Two additional supplied count-replay carriers (DeepSeek-R1 at $M = 2$ and Qwen3-4B at $N = 8$) are setting-specific descriptive comparisons, not a policy-class comparison or a general allocation law (Section 6).
5. **Honest boundaries.** The public carrier gives per-task pass *counts*, not ordered rollouts; it supports a random-prefix (count-exchangeable) analysis only as a model assumption and cannot validate ordered online deployment. We therefore bound the failure domain under three pre-fixed synthetic ordered-drift mechanisms; every displayed margin result is synthetic, exploratory, and nonconfirmatory (Section 7).

2 Related work

Adaptive sampling for voting. ASC (Aggarwal et al., 2023) adapts a per-problem sample count using a lightweight stopping criterion, and ESC (Li et al., 2024) proposes early stopping for self-consistency. We do not establish a finite-sample disagreement-with-full-vote guarantee for either; on our supplied heterogeneous replay carrier, the instantiated ESC-style rule exceeds 0.05 (Table 1). MARS (Chen et al., 2026) is a parallel trace-stopping method that bounds future vote movement; our work is instead a sequential count-exchangeable replay certificate. Best-of-n and budget-forcing methods (Snell et al., 2025) tune compute without stopping certificates.

Conformal and anytime inference. Conformal prediction (Vovk et al., 2005) gives marginal coverage under exchangeability; confidence sequences (Howard et al., 2021) give anytime-valid bounds for i.i.d. means. Our oracle object is neither: under the true $H_\star$, $c_{H_\star}$ is an *exact posterior flip probability*, valid conditionally per prefix. The implemented fitted score is instead screened through the finite FIT-frozen CAL empirical-Bernstein/Bonferroni construction of Theorem 2; it is not claimed conditionally exact per prefix.

Bayesian stopping and finite-population inference. Rossell and Müller (Rossell & Müller, 2013) and Novikov (Novikov, 2010) are Bayesian decision-theoretic sequential-stopping antecedents; neither is a held-out task-level calibration result for a fitted score under this replay endpoint. For the sampling model, Waudby-Smith and Ramdas (Waudby-Smith & Ramdas, 2020) are the closest finite-population without-replacement precedent: their binary prefix counts are hypergeometric and their confidence sequences remain frequentist under a working prior. A plausible-count set wholly on one side of the full-N majority threshold induces a direct WoR-CS majority-stopper under the same random-prefix model; it is not compared here. Thus our contribution is only model-assisted FIT/CAL screening of a finite frozen plug-in-score family by exact task-level replay loss, with no new CS or efficiency-separation claim.

Dependent sampling. Design-effect corrections (Kish, 1965) shrink effective sample size by an intraclass correlation; we show pooled correlation is the wrong functional here (Section 3). Beta-binomial and correlated-Bernoulli models (Skellam, 1948) capture overdispersion but target estimation, not stopping certificates for the flip event.

Feature boundary. Table 2 makes the source-verified distinction compact: the proposed object is a specified count-exchangeable full-vote *replay* endpoint with an oracle conditional object and a distinct fitted-score CAL guarantee. It is not a general ordered-online early-stopping, conformal, or anytime-inference guarantee.

3 Model: count-exchangeable replay

A task's N rollout outcomes are summarized by the pass count $K \in \{0, \dots, N\}$. Across tasks, $K \sim H_\star$, the *true* mixture capturing both task heterogeneity and any within-task correlation: the probability a fresh rollout passes, marginally, is $\mathbb{E}_{H_\star}[K]/N$, and the joint law of the N outcomes is exchangeable *within* a task given its realized count. Analysis model (*replay*): the stopper observes a uniform random prefix drawn without replacement from the task's N outcomes — i.e., the first k draws are exchangeable with the remaining $N - k$. Section 7 stress-tests departures from this model (ordered drift); our validity claims are stated relative to it, and the carrier limitation (counts only, no order) is made explicit in Section 8.

Let $\mathrm{MV}(k) = \mathbf{1}\{x_k > k/2\}$ with ties assigned to the negative side, and let $\mathrm{MV}(N) = \mathbf{1}\{K > N/2\}$ be the full-N *binary pass-count replay decision*. It is not an aggregation of delivered textual answers or an actual delivered answer-majority: the replay model contains only binary pass/fail outcomes. The *flip event* of a stopping rule τ is $\{\mathrm{MV}(\tau) \neq \mathrm{MV}(N)\}$.

Remark 1 (Why the mixture, not the correlation). On τ^2-bench carriers (6 public task suites, 4 rollouts/task) we measured pooled within-task reward correlation $\hat{\rho} = 0.47$–0.59, but a task-demeaned (fixed-effect) estimate gives $\hat{\rho}_{\mathrm{FE}} \approx -1/3 \approx -1/(N-1)$. This fixed-effect value is a mechanical demeaning identity: within each fixed-N task the residuals sum to zero, so it cannot identify the presence or dominance of latent within-task dependence. On OpenMathReasoning the per-task pass-rate variance exceeds the binomial baseline by $34\times$. Under the stated count-exchangeable replay model, the flip probability is the count-mixture functional $\sum_K H[K] f_K(k)$ and is driven by mass near $K \approx N/2$; pooled or demeaned second moments do not identify that functional or its heterogeneity/dependence source.

4 METHOD: THE MIXTURE-POSTERIOR STOPPER

Exact per-task flip probability. Given count K, the prefix count x_k is hypergeometric $\mathrm{HG}(N, K, k)$, hence

$$f_K(k) \;:=\; \mathbb{P}\big(\mathrm{MV}(k) \neq \mathrm{MV}(N) \,\big|\, K\big) = \sum_x \mathrm{HG}(N, K, k, x)\, \big|\mathrm{side}(x, k) - \mathrm{side}(K, N)\big|. \quad (1)$$

Lemma 1 (Exactness and monotonicity)**.** (1) *is an identity (no approximation). For every fixed K, $f_K(k)$ is nonincreasing on the odd early-budget grid $\mathcal{O}_N = \{k \in \{1, \ldots, N-1\} : k \text{ odd}\}$. Hence, for any mixture H, $\mathbb{E}_H[f_K(k)]$ is nonincreasing. For a selected odd subgrid $\mathcal{O} \subseteq \mathcal{O}_N$, define $\mathcal{F}_H(\alpha) = \{k \in \mathcal{O} : \mathbb{E}_H[f_K(k)] \leq \alpha\}$. Only when $\mathcal{F}_H(\alpha) \neq \varnothing$ is $k_H^{\mathrm{odd}}(\alpha) = \min \mathcal{F}_H(\alpha)$ defined and binary-searchable; otherwise no odd early budget meets the criterion and the protocol's separate full-N fallback is not called an odd k_H^{odd}.*

Proof. The identity is a definition expansion. Monotonicity follows from a coupling of the prefixes at k and $k{+}2$ on a common ordered draw. The proof first handles the support-boundary case where the downward boundary atom has probability zero directly; only otherwise does it divide to obtain the ratio $(K-t)/(N-K-t) \geq 0$ for $K > N/2$. The complementary case follows by pass/fail reflection for $K < N/2$; the even-N center case $K = N/2$ is separately constant at $1/2$ for odd k (Appendix A). An exact-rational CPU audit checks every admissible (N, K, k) for $2 \leq N \leq 64$, including $(32, 17, 29)$, the center case, and the no-feasible-odd-budget full-N fallback semantics. □

Oracle conditional certificate (true $H_\star$ only). Given the true count law $H_\star$ and an observed prefix (x, k), define the oracle posterior flip probability

$$c_{H_\star}(x, k) \;:=\; \mathbb{P}_{H_\star}\big(\mathrm{side}(K, N) \neq \mathrm{side}(x, k) \,\big|\, x, k\big), \qquad \text{posterior} \propto H_\star[K] \cdot \mathrm{HG}(N, K, k, x). \quad (2)$$

We define the terminal oracle certificate by $c_{H_\star}(x, N) := 0$. At a posterior-supported terminal state, the hypergeometric likelihood has support only on $K = x_N$, so this convention equals the conditional flip probability: $c_{H_\star}(x_N, N) = 0$. It also gives the correct endpoint value for a zero-prior-mass row of a frozen fitted-score table, because the observed total count already fixes the binary full-N replay decision.

Theorem 1 (Oracle conditional identity and adaptive bound)**.** *Let τ be any stopping time measurable w.r.t. the prefix filtration that stops only at states with $c_{H_\star}(x_\tau, \tau) \leq \alpha$. The terminal state $k = N$ satisfies this same rule because $c_{H_\star}(x_N, N) = 0$. Under the true law $H_\star$, $\mathbb{P}_{H_\star}(\mathrm{MV}(\tau) \neq \mathrm{MV}(N)) = \mathbb{E}_{H_\star}[c_{H_\star}(x_\tau, \tau)] \leq \alpha$, with equality in the identity and no union bound over stopping times.*

Proof. $\mathbb{P}_{H_\star}(\text{flip}) = \mathbb{E}_{H_\star}[\mathbb{P}_{H_\star}(\text{flip} \mid \text{prefix path})]$. For any path stopped at (x, k), $\mathbb{P}_{H_\star}(\mathrm{MV}(N) \neq \mathrm{side}(x, k) \mid \text{path}) = c_{H_\star}(x, k)$ since the posterior is exactly the conditional law of K under $H_\star$ and $\mathrm{MV}(N)$ is a function of K. At $\tau = N$, $x_N = K$ and hence this conditional probability is $c_{H_\star}(x_N, N) = 0$; at every earlier stopped state it is at most α by the rule. Thus the integrand is bounded pointwise by α, and taking expectations gives the claim. □

Fitted implementation (BAYES-H). The implementation freezes $\widehat{H}_{\mathrm{FIT}}$ before CAL and thresholds the *fitted stopping score*

$$s_{\widehat{H}_{\mathrm{FIT}}}(x, k) := \mathbb{P}_{\widehat{H}_{\mathrm{FIT}}}\big(\mathrm{side}(K, N) \neq \mathrm{side}(x, k) \mid x, k\big).$$

It stops at $\widehat{\tau} = \min\{k \geq 3 : s_{\widehat{H}_{\mathrm{FIT}}}(x_k, k) \leq \alpha\}$. This score is a plug-in posterior calculation, not a claim that $s_{\widehat{H}_{\mathrm{FIT}}}(x, k) = \mathbb{P}_{H_\star}(\text{flip} \mid x, k)$ unless $\widehat{H}_{\mathrm{FIT}} = H_\star$. Its terminal value is set to zero because $x_N = K$, but that endpoint fact does not repair the non-oracle conditional identity at earlier prefixes. Once the fitted rule is frozen, its exact per-task replay-flip probability and expected stopping time under any true count K are computable by DP over prefix states (k, x) (Appendix B). The operational guarantee for this fitted rule is Theorem 2, not Theorem 1.

Theorem 2 (CAL-screened marginal replay guarantee for a FIT-frozen rule family)**.** *Let FIT and CAL be independent i.i.d. task samples from a common count-exchangeable replay population with true count law $H_\star$. Conditional on FIT, freeze $\widehat{H}_{\mathrm{FIT}}$, a finite pre-declared family $\mathcal{R}_{\mathrm{FIT}}$ (including all fitted-score tables), and fix $r \in \mathcal{R}_{\mathrm{FIT}}$. Let $g_r(K)$ be the exact per-task replay flip probability of this* fixed fitted rule, *and let $\bar{g}_{\mathrm{CAL},r}$, $s^2_{\mathrm{CAL},r}$ be the mean and variance of $g_r(K)$ over the m CAL tasks. For a per-rule failure allocation δ_r, with probability $1-\delta_r$ over CAL conditional on FIT,*

$$\mathbb{P}_{\textit{replay},H_\star}(\mathrm{flip}) \ \le\ \bar{g}_{\mathrm{CAL},r} + \sqrt{\frac{2s^2_{\mathrm{CAL},r}\log(4/\delta_r)}{m}} + \frac{7\log(4/\delta_r)}{3(m-1)},$$

where the first term is $\bar{g}_{\mathrm{CAL},r}$. This is the empirical-Bernstein inequality (Maurer & Pontil, 2009) for bounded i.i.d. per-task flips $g_r \in [0,1]$. Setting $\delta_r = \delta/J$ and applying Bonferroni gives a simultaneous CAL selection guarantee over J frozen candidates (Section 5). This is a marginal replay statement for the CAL-screened FIT-frozen rule; it does not assert a per-problem oracle conditional identity for $s_{\widehat{H}_{\mathrm{FIT}}}$.

Remark 2 (No Jensen gap). For a fixed fitted rule, the population replay flip rate is a *linear* functional of $H_\star$: $\mathbb{P}_{H_\star}(\mathrm{flip}) = \sum_K H_\star[K]\, g(K)$. Hence the CAL bound applies to a bounded per-task quantity; it does not make the fitted score an oracle conditional certificate.

Proposition 1 (Exact deterministic TV sensitivity of a frozen rule)**.** *Fix the rule (fitted-score tables frozen from $\widehat{H}$) and let $g(K)$ be its exact per-task flip probability. Over the total-variation ball $B_R = \{q : q \ge 0, \sum_K q_K = 1, \|q - \widehat{H}\|_1 \le 2R\}$,*

$$V(R) \ :=\ \max_{q \in B_R}\ \textstyle\sum_K q_K\, g(K)$$

is a capacity-constrained linear program whose optimum is a two-sided greedy: add mass to atoms in descending g (capped by $q_K \le 1$) and drain mass from atoms in ascending g (capped by $\widehat{H}_K$). It is computable in $O(N \log N)$ and is nondecreasing, piecewise linear in R; it is not generally globally affine because donor and recipient capacities create breakpoints. Any two simplex points have ℓ_1 distance at most 2, so B_1 is the full simplex; the linear objective is maximized at a vertex and therefore $V(1) = \max_K g(K)$. For the positive radii and realized capacities checked in the supplied LP grids, it is strictly below the 1-Lipschitz envelope $\mathrm{flip}(\widehat{H}) + R$*; no universal strict-refinement claim is made. The set $\mathcal{R}_\alpha = \{R \in [0,1] : V(R) \le \alpha\}$ is nonempty exactly when $V(0) \le \alpha$. Only in that case define the* critical radius *$\tau^*(\alpha) = \max \mathcal{R}_\alpha$; if $V(0) > \alpha$, record the explicit infeasible sentinel* `infeasible` *rather than the numeric value 0. It becomes a population certificate only after separately proving that an unknown target mixture lies in B_R, which current artifacts do not do.*

The stored OMR, RLVE, and OpenR1 curves in Appendix E are deterministic visualizations of this assumed-ball calculation for their respective fitted rules. They are retained only as sensitivity diagnostics: they neither establish target-population containment nor justify a cross-carrier mechanism, a shift-robustness ranking, or a prescription for choosing α.

Remark 3 (Reported fitted-score family for CAL screening). The reported protocol uses only the finite, pre-declared $J = 64$ family in Section 5: 60 fixed-k candidates and four BAYES-H levels. Theorem 2's Bonferroni statement is only for that FIT-frozen family. A stored breakpoint enumeration over other α values is not a selection family for the reported table and supplies no additional operational guarantee.

5 PROTOCOL: FIT/CAL/TEST WITH A LEGITIMATE SELECTION FAMILY

Split and status. 11,607 problems are partitioned once (seeded) into FIT (4000), CAL (4000), TEST (3607). The population statement in Theorem 2 assumes tasks are i.i.d. draws from one replay population and that FIT and CAL are independent. FIT estimates $\widehat{H}_{\mathrm{FIT}}$ and freezes the BAYES-H fitted-score tables and the pre-declared family before CAL is inspected. CAL selects rules from that frozen family: fixed-k candidates $k \in \{3, 5, \ldots, 31\} \times \alpha \in \{0.10, 0.05, 0.02, 0.01\}$ (60 candidates) and BAYES-H at the same 4 levels, with simultaneous empirical-Bernstein UCBs at per-rule $\delta_r = 0.05/64$ and Bonferroni over $J = 64$ FIT-frozen candidates. For each named fixed-budget selector and level, let $\mathcal{F}_{\mathrm{CAL}}(\alpha)$ be its odd budgets in $\{3, 5, \ldots, 31\}$ whose declared CAL UCB is $\le \alpha$. $k^*(\alpha)$ denotes the smallest element only when this set is nonempty; otherwise the table says "no CAL-screened early budget" and that fixed policy uses the deterministic FULL-N

Table 1: Fixed-benchmark descriptive TEST readout ($n = 3607$ problems, $N = 32$). "CAL-selected" means that the FIT-frozen candidate passed its Bonferroni CAL empirical-Bernstein UCB criterion; it is not a TEST guarantee or oracle conditional certificate. "No CAL screen" marks a reference or heuristic. Replay count reduction $= 1 - \bar{k}/32$. At $\alpha = 0.05$ and 0.02, this TEST table reports larger replay count reduction for BAYES-H than the named fixed rows; at $\alpha = 0.10$, FIXED-EB reports 84.4% versus BAYES-H 84.0%. WINDOW3 has the largest reported reduction but exceeds the replay-flip target at $\alpha \leq 0.05$.

Rule	α	replay full-vote flip	mean k	count reduction
Status. CAL-selected rows passed a FIT-frozen Bonferroni CAL EB UCB; TEST values are descriptive. A "no CAL-screened budget" row uses the separate FULL-32 fallback (not a CAL-selected member of J). FULL is a reference; WINDOW3 has no CAL screen.				
FULL-32	—	0	32.0	0
WINDOW3 (ESC-style)	—	**0.0577**	4.4	86.2%
FIXED-HOEF k^*=7	0.10	0.0645	7.0	78.1%
FIXED-EB k^*=5	0.10	0.0794	5.0	84.4%
BAYES-H	0.10	**0.0370**	5.1	84.0%
FIXED-HOEF k^*=27	0.05	0.0156	27.0	15.6%
FIXED-EB k^*=17	0.05	0.0324	17.0	46.9%
BAYES-H	0.05	**0.0249**	6.1	**80.9%**
FIXED-HOEF	0.02		*no CAL-screened budget*	
FIXED-EB k^*=31	0.02	0.0083	31.0	3.1%
BAYES-H	0.02	**0.0091**	8.5	**73.5%**

fallback (zero flip, mean budget N), outside $J = 64$ and not CAL-selected. BAYES-H is accepted at level α iff its CAL EB UCB is $\leq \alpha$. These are marginal replay UCBs for fitted frozen rules, not per-problem oracle conditional certificates. TEST is read exactly once, not used for selection, and is a fixed-benchmark descriptive readout rather than a second population guarantee. All evaluation uses the exact per-task flip functions ((1) for fixed-k; the DP of Appendix B for BAYES-H), so TEST numbers are exact replay quantities, not MC estimates; the DP itself is cross-checked against replay MC (max deviation 0.043 within $3\sigma_{\mathrm{MC}} = 0.075$).

Baselines. FULL-N; FIXED-HOEF (fixed budget selected with distribution-free Hoeffding UCBs, a named model-free comparator); FIXED-EB (fixed budget with the same mixture-aware empirical-Bernstein machinery as BAYES-H — isolates the value of adaptivity); BAYES-UNIF (uniform prior; no fitted-score CAL guarantee when the prior is wrong); WINDOW3 (ESC-style: stop when 3 consecutive samples agree with the current majority — no CAL-screened tuning rule).

6 Experiments

Carrier. OpenMathReasoning CoT shard 0 (Moshkov et al., 2025) (CC-BY-4.0): the reviewer-safe derived manifest records 22,230 source rows, 12,423 valid count rows, and 11,607 first-occurrence deduplicated problems. For each retained problem, the frozen derivation maps `pass_rate_72b_tir` to $K =$ `int(round(float(pass_rate_72b_tir) * 32))`. The manifest supports byte-exact replay of the main JSON conditional on that derived data; it does not expose the raw parquet, raw evaluator, or answer-extraction path, and is not a clean-room reconstruction. Pass rates are strongly bimodal: 45% of problems have $p \leq 0.1$ or $p \geq 0.9$; 11.6% sit in $[0.4, 0.6]$. This carrier is a count-only carrier used for the *count-exchangeable replay* analysis: it supplies per-task counts but no rollout order, so the random-prefix assumption is a modeling condition rather than an observed chronology (Section 7).

Main table. Table 1 reports the single TEST readout (Bonferroni CI radius 0.0286 across the 9 reported rules). In this section, "valid" means replay full-vote flip $\leq \alpha$ and TEST is descriptive (selection was certified on CAL), not gold correctness or an operational cost. Table 5 compactly maps the similarly descriptive TEST outcomes.

Fair comparison. Against the named distribution-free baseline (FIXED-HOEF) at the same α, BAYES-H's paired per-problem replay-count-reduction gap is $+0.058$ (CI radius 0.029) at $\alpha = 0.10$ and $+0.652 \pm 0.029$ at $\alpha = 0.05$ — a $5.2\times$ improvement over the baseline's 15.6% count reduction; at $\alpha = 0.02$ the baseline has *no* CAL-screened budget while BAYES-H still has a 73.5% reduction. Against the named mixture-aware fixed budget (FIXED-EB), this fitted-score rule reports $+34$ points of replay count reduction at $\alpha = 0.05$ (80.9% vs 46.9%) and $+70$ points at $\alpha = 0.02$ — the adaptive gap in this TEST readout grows as α tightens. This descriptive pattern is consistent with the fitted score stopping earlier on extreme-K problems than on the 11.6% mid-difficulty mass; it is not an oracle conditional claim for $\widehat{H}_{\mathrm{FIT}}$.

Cross-source transfer. Splitting CAL/TEST across the four AoPS problem sources (train on three, test on the fourth), the supplied BAYES-H readouts have replay flip 0.021–$0.033 < \alpha = 0.05$ and count reduction 79–83%. These are descriptive source-split replay readouts, not a population-transfer, oracle-conditional, or ordered-online transfer guarantee.

Second-shard replication and prior-mismatch transfer. We replicate the entire chain on a disjoint carrier, OpenMathReasoning CoT shard 1 (11,454 problems, same N=32 pass counts; SHA-256 pinned), whose mixture structure is near-identical (heterogeneity excess 0.125 vs 0.129; 44.4% extreme-p mass). Two runs share the CAL/TEST problems of shard 1 and differ only in the prior's provenance. *(i) Within-shard*: the prior is fit on shard 1's own FIT split; every qualitative replay conclusion reproduces — BAYES-H has 80.6% replay count reduction at realized flip 0.025 ($\alpha = 0.05$) and 73.1% at 0.009 ($\alpha = 0.02$), vs 15.6% and *no CAL-screened budget* for FIXED-HOEF. *(ii) Cross-shard prior transfer*: the prior is fit on shard 0 FIT (fitted-mixture TV distance to the shard 1 fit $= 0.042$) and the fitted-score candidate is CAL-screened and read on shard 1. All numbers move by ≤ 0.4 points (count reduction $80.6\%/73.2\%$; flips $0.025/0.009$). This is a small movement in one supplied count-replay transfer and a deterministic fitted-mixture sensitivity diagnostic, not a statistical population-transfer certificate or a demonstration of robustness to naturally ordered online rollouts (Section 3).

Second model–carrier pair: OpenR1-Math-220k (DeepSeek-R1, $M = 2$). The supplied raw table contains 9,374 unique problems (HF snapshot `e4e141ec`). The frozen inclusion predicate hashes each generated string within a problem, removes duplicate rollouts, and retains a problem *only if exactly two deduplicated rollouts remain*; 8,853 problems meet that predicate. A seeded shuffle then makes the analysis split FIT/CAL/TEST $3000/3000/2853$. Thus 9,374 is the raw-unique count and 8,853 is the analyzed count; they are not interchangeable. Under the replay model, the full vote is the 2-vote majority (ties use a pre-drawn fair coin), the only stoppable prefix is $k = 1$, and the fitted mixture has the three identifiable atoms $p \in \{0, \frac{1}{2}, 1\}$.

For the named fitted-score candidates, the descriptive TEST readout is 50% replay count reduction and flip 0.0592 at $\alpha = 0.10$, 32.1% and 0.0294 at $\alpha = 0.05$, and a never-stop row at $\alpha = 0.02$. The CAL EB UCBs are 0.0707, 0.0383, and 0.0047, respectively, under the frozen five-rule Bonferroni family. These values describe the evaluated candidates; they do not establish a policy-class comparison or a general allocation law. They also do not test chronological online deployment, gold correctness, or operational cost.

Third model–carrier pair: RLVE (Qwen3-4B, $N = 8$). The two carriers above close the extremes (N=32 rich-prefix vs M=2 near-prefix-free). To probe the intermediate regime under a third model family and a different task family, we run the identical chain on RLVE teacher rollouts (Qwen3-4B-Thinking-2507, pass@8 on 9,000 verifiable counting/combinatorics environments; Apache-2.0; HF snapshot `eaeec946`). Success is the offline Gym-verifier reward being positive (a small mass of fractional partial credit is counted as failure). The mixture is again strongly U-shaped (42% extreme problems, heterogeneity excess 0.100 over the binomial baseline). Results (FIT/CAL/TEST 3000/3000/3000): *(i)* BAYES-H has replay count reduction $57.6\%/50.9\%/48.2\%/45.4\%$ at $\alpha = 0.10/0.05/0.02/0.01$ with realized flips $0.031/0.010/0.005/0.002$ — conservative at every level; *(ii)* the fitted-score candidate's CAL EB UCB is at most α at every reported level down to $\alpha = 0.01$, while *no* distribution-free budget is CAL-screened below $\alpha = 0.10$ (FIXED-EB and FIXED-Hoeffding both null at $\alpha \leq 0.05$) — at $\alpha = 0.10$ the paired replay-count-reduction gap over FIXED-Hoeffding is $+0.20$ (CI radius 0.03, significant); *(iii)* the exact DP prediction matches a 3,000-draw replay Monte-Carlo on held-out problems (flip within 0.004, mean k within 0.04). Two descriptive diagnostics: the recorded sample-id success slope is flat (pooled slope -7.8×10^{-5}/trial; per-position rates 0.316–0.326), consistent with near-exchangeable sampling in this carrier; and the reported replay-count ranges differ across the three supplied carrier settings (81% at N=32 with 15 candidate budgets, 45–58% at N=8 with 3, 32–50% at M=2 with 1; all readouts are collected in Appendix F). This comparison is descriptive only: it neither establishes a rule beyond its reported settings nor audits an online policy or makes the count-replay guarantee valid for deployment order.

Why the window heuristic fails. On mid-difficulty problems ($K \approx 16$), three consecutive agreements with the current majority is a *common* event under near-coin-flip draws, not evidence the vote has settled: WINDOW3 realizes flip 0.058 at mean $k = 4.4$. Its flip rate is nearly flat in α (it has no CAL-screened tuning rule): it can only be tuned by window length, and its certified population UCB is 0.061 (`adaptive_sweep_r468_result.json`), so it barely misses even the loosest tested level $\alpha = 0.10$ on the full OMR test pool.

7 Synthetic ordered drift: exploratory stress diagnostic

The replay model assumes prefixes are exchangeable with the held remainder. Because the carrier contains no observed chronology, we apply three *pre-fixed synthetic* drift mechanisms to its true counts ($\delta \in \{0, 0.05, 0.10, 0.15, 0.20\}$, 2000 TEST problems, 500 orders each): **E1** (front-load: the first k draws are enriched/depleted in passes by count $D = \lceil 2\delta \min(K, N-K) \rceil$), **E2** (linear per-draw pass-rate trend $p_j = K/N + \delta(2j/(N-1)-1)$), **E3** (adversarial block: the last $\lceil \delta N \rceil$ draws drift toward the minority side).

Findings (full tables in Appendix D) are a specified synthetic stress diagnostic, not evidence about natural order. All methods exceed the target under E1 by $\delta = 0.10$–0.15 in this harness. The grid also has non-uniform outcomes: under E2 at $\alpha = 0.10$, BAYES-H's replay flip is 0.083 at $\delta = 0.20$, whereas under E3 at $\alpha = 0.05$ and $\delta = 0.15$, BAYES-H is invalid (0.052) while FIXED-EB remains valid (0.047). Thus the grid supplies no global robustness ranking.

Every margin-exploration result is *synthetic, exploratory, and nonconfirmatory*. No frozen margin-selection record is bundled. For example, in the E3 $\alpha = 0.05$ stress harness, the explored $\gamma = 0.025$ row has flip 0.0495 and replay count reduction 77.2% at $\delta = 0.15$, versus 81.0% for the unmargined row; at $\delta = 0.20$, even $\gamma = 0.03$ is invalid. Under E2, $\gamma = 0.03$ has flip 0.0491 and count reduction 71.8% at $\delta = 0.20$. These values are neither CAL-selected nor confirmatory and do not certify an online policy. The drift budget δ could be estimated only if ordered rollouts were retained; on this count-only carrier it is assumed.

8 Limitations

(i) The public carrier provides pass *counts*, not ordered rollouts. The oracle identity in Theorem 1 is exact only under the true count-exchangeable law $H_\star$. The implemented $s_{\widehat{H}_{\mathrm{FIT}}}$ is a fitted stopping score, not that conditional identity; its operational statement is only Theorem 2's FIT-frozen CAL-screened marginal replay guarantee. The synthetic stress mechanisms of Section 7 are specified departures from the replay model, not natural-order evidence. No supplied run compares the stopped answer with gold correctness or records generated tokens, latency, cancellation acknowledgement, or post-cancellation work; full-vote agreement and $1 - \bar{k}/N$ cannot substitute for those endpoints.

(ii) The mixture prior is estimated from the 4000 FIT problems; Theorem 2 assumes CAL is an independent sample from the same replay population. The numeric critical radii of Proposition 1, reported only when $V(0) \leq \alpha$, are deterministic fitted-rule sensitivities over an assumed fitted-mixture TV ball ($\tau^* = 0.024$–0.157 across the named levels and shards); they do not establish target-population containment, natural-shift robustness, or a policy-selection prescription. Appendix E(g) records only an oracle K-wise profile-capping sensitivity: its subset is selected separately after conditioning on final count K, which is unavailable at a prefix. It is neither a common prefix-state rule nor an instance of Theorem 1, and supports no stopping-policy, cost, or repair claim.

(iii) Lemma 1's monotonicity is proved in closed form (Appendix A) and independently checked exhaustively at $N = 32$. (iv) The additional OMR shard, OpenR1, and RLVE results are supplied setting-specific replay readouts; they do not establish generality or an allocation law. OpenR1 in particular distinguishes 9,374 raw unique problems from the 8,853 exactly-two-deduplicated-rollout analysis population and its $3000/3000/2853$ split. (v) At $k = N$, the observed pass count equals K, so the binary pass-count replay decision agrees with itself and both the oracle score and fitted terminal score are zero. Terminal stops add zero replay flip for every true count; that endpoint fact does not identify fitted earlier prefix scores with oracle conditional probabilities.

9 Conclusion

Under the count-exchangeable replay model, the true law $H_\star$ yields an exact oracle conditional identity and tower-property theorem. BAYES-H uses a different fitted score, whose operational claim is only the FIT-frozen, Bonferroni CAL-screened marginal replay guarantee of Theorem 2. The supplied OMR descriptive TEST readout reports 80.9% replay rollout-count reduction at replay flip 0.0249 for $\alpha = 0.05$, alongside FIXED-HOEF 15.6% and FIXED-EB 46.9% rows. The drift and margin results remain synthetic, exploratory, and nonconfirmatory. Natural online order, gold correctness, and actual token/latency/cancellation costs remain open.

Reproducibility statement

The count-replay analyses, assumptions, split protocol, and complete proofs are specified in Sections 3–5 and the appendix. The anonymous reviewer-safe bundle includes a derived count-level manifest and a standard-library verifier that reproduces the main result JSON byte-for-byte (SHA-256 `b114c72d...c8a5f7`) conditional on that manifest. It does not reconstruct the manifest from the omitted raw parquet, and the recovered secondary runners are preserved for provenance rather than locally rerun. The absence of ordered online telemetry and other claim boundaries are stated in Section 8.

AI-use statement

Generative AI tools were used to assist manuscript-language revision, evidence and citation auditing, build checks, and internal pre-submission review, including the Codex multi-agent revision and audit documented for this version. Figure 1 is a code-native Matplotlib rendering from hash-pinned frozen JSON inputs, rather than a generative image. These tools did not generate, execute, or replace the reported scientific experiments, their quantitative result artifacts or values, or the final technical judgments; the authors reviewed the changes and remain responsible for the paper.

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A PROOFS

A.1 MONOTONICITY OF THE FIXED-k FLIP PROBABILITY (LEMMA 1)

Recall

$$f_K(k) = \mathbb{P}\big(\mathrm{MV}(k) \neq \mathrm{MV}(N) \,\big|\, K\big), \qquad x_k \sim \mathrm{HG}(N, K, k),$$

with the negative side winning ties ($\mathrm{MV}(k) = 1$ iff $x_k > k/2$). We prove that $f_K(k)$ is nonincreasing along the odd numbers $k \in \{1, 3, 5, \ldots, N-1\}$ for every fixed K; this suffices for the paper, since its early-budget candidates are odd. For a selected odd subgrid, $k_H^{\mathrm{odd}}(\alpha)$ is defined only when its feasible set $\mathcal{F}_H(\alpha) = \{k : \mathbb{E}_H[f_K(k)] \leq \alpha\}$ is nonempty; otherwise the protocol uses its separate full-N fallback rather than calling it an odd certificate.

Setup and case split. $\mathrm{MV}(k) \neq \mathrm{MV}(N)$ can happen in only two ways:

- **Case U (upset).** $K < N/2$ (full vote negative) and $x_k > k/2$ (prefix votes positive).
- **Case R (reversal).** $K > N/2$ (full vote positive) and $x_k \leq k/2$ (prefix votes negative).

When N is even there is also the center case $K = N/2$, handled separately below. Write $f_K(k) = U_K(k) + R_K(k)$ with $U_K(k) = \mathbb{P}(x_k > k/2)\,\mathbf{1}\{K < N/2\}$ and $R_K(k) = \mathbb{P}(x_k \leq k/2)\,\mathbf{1}\{K > N/2\}$. At most one of the two terms is nonzero away from the center.

The reflection identity. Swapping the labels "pass"/"fail" of all N rollouts maps the law $\mathrm{HG}(N, K, k)$ of x_k onto the law of $k - x_k$ under count $N - K$, i.e. $\mathbb{P}_K(x_k > s) = \mathbb{P}_{N-K}(x_k < k - s)$. Hence, for odd k,

$$U_K(k) = \mathbb{P}_K(x_k > k/2) = \mathbb{P}_{N-K}(x_k < k/2) = R_{N-K}(k) \qquad (K < N/2),$$

since k odd makes $k/2$ a half-integer, so the strict and weak inequalities coincide. Every upset probability thus equals a reversal probability at the reflected count, and it suffices to prove

$$R_K(k+2) \leq R_K(k) \qquad \text{for all } K > N/2 \text{ and odd } k, \tag{3}$$

which covers Case R directly and Case U by reflection (the same display applied at count $N - K$).

Even-N center case. If N is even and $K = N/2$, symmetry under $x \mapsto k - x$ and the absence of an odd-k tie give $\mathbb{P}(x_k > k/2) = \mathbb{P}(x_k < k/2) = 1/2$. The full vote is negative, so $f_{N/2}(k) = 1/2$ for every odd k: it is constant. This is why reflection is restricted to $K < N/2$.

Coupling construction. Fix $K > N/2$ and odd $k \le N - 3$. We couple the two hypergeometric draws on a single probability space. Let the population of N rollouts consist of K passes and $N-K$ fails. Draw an *ordered* sample of size $k + 2$ uniformly without replacement and split it as

$$\underbrace{X_1, \ldots, X_k}_{\text{prefix } k} \;\Big|\; \underbrace{X_{k+1}, X_{k+2}}_{\text{extra pair } (A,B)}.$$

The marginal law of $(X_1, \ldots, X_k)$ is exactly $\mathrm{HG}(N, K, k)$, and $x_{k+2} = x_k + A + B$ has law $\mathrm{HG}(N, K, k+2)$. All statements below are about this joint space, so they hold for the two marginals simultaneously.

Let $t = k/2 + 1/2$ be the (integer) flip threshold: the prefix at k votes negative iff $x_k \le t - 1$; the prefix at $k + 2$ votes negative iff $x_{k+2} \le t$. Partition the event $\{x_k = s\}$ by the value of the pair $A + B \in \{0, 1, 2\}$:

$$\begin{aligned}
E_0 &= \{x_k = t-1,\ A+B = 0\}, && \text{flip at } k,\ \text{flip at } k+2,\\
E_1 &= \{x_k = t-1,\ A+B = 1\}, && \text{flip at } k,\ \text{flip at } k+2,\\
E_2 &= \{x_k = t-1,\ A+B = 2\}, && \text{flip at } k,\ \text{no flip at } k+2,\\
F_0 &= \{x_k = t,\quad A+B = 0\}, && \text{no flip at } k,\ \text{flip at } k+2,\\
F_1 &= \{x_k = t,\quad A+B = 1\}, && \text{no flip at } k,\ \text{no flip at } k+2.
\end{aligned}$$

All other states agree on both sides (e.g. $x_k \le t-2$ flips under both, since $A + B \le 2$; $x_k \ge t+1$ flips under neither). Therefore

$$R_K(k) - R_K(k+2) = \mathbb{P}(E_2) - \mathbb{P}(F_0), \tag{4}$$

and it remains to show $\mathbb{P}(E_2) \ge \mathbb{P}(F_0)$.

Comparison of the two boundary atoms, including support boundaries. Both atoms factor as (prefix probability) $\times$ (pair probability). We first split off the case $\mathbb{P}(F_0) = 0$. Then (4) gives $R_K(k) - R_K(k+2) = \mathbb{P}(E_2) \ge 0$ directly; in particular, no ratio is formed when a binomial factor in F_0 vanishes (including a zero remaining-fail count or an infeasible $x_k = t$ atom). It remains to consider $\mathbb{P}(F_0) > 0$. This support condition makes every denominator below strictly positive and makes the ratio algebra legitimate. With $k = 2t - 1$, first,

$$\mathbb{P}(x_k = t-1) = \frac{\binom{K}{t-1}\binom{N-K}{t}}{\binom{N}{k}}, \qquad \mathbb{P}(x_k = t) = \frac{\binom{K}{t}\binom{N-K}{t-1}}{\binom{N}{k}},$$

since $k - (t-1) = t$ and $k - t = t - 1$. Given $x_k = t - 1$, the remainder holds $K - t + 1$ passes out of $N - k$, so $\mathbb{P}(A + B = 2 \mid x_k = t-1) = \binom{K-t+1}{2}/\binom{N-k}{2}$; given $x_k = t$, the remainder holds $N - K - t + 1$ fails, so $\mathbb{P}(A + B = 0 \mid x_k = t) = \binom{N-K-t+1}{2}/\binom{N-k}{2}$. The denominators $\binom{N}{k}\binom{N-k}{2}$ are common to both atoms. Under $\mathbb{P}(F_0) > 0$, hence

$$\frac{\mathbb{P}(E_2)}{\mathbb{P}(F_0)} = \frac{\binom{K}{t-1}\binom{N-K}{t}\binom{K-t+1}{2}}{\binom{K}{t}\binom{N-K}{t-1}\binom{N-K-t+1}{2}} = \frac{K-t}{N-K-t},$$

where the last equality uses the three elementary ratios $\binom{K}{t-1}/\binom{K}{t} = \frac{t}{K-t+1}$, $\binom{N-K}{t}/\binom{N-K}{t-1} = \frac{N-K-t+1}{t}$, and $\binom{K-t+1}{2}/\binom{N-K-t+1}{2} = \frac{(K-t+1)(K-t)}{(N-K-t+1)(N-K-t)}$, whose product telescopes to the displayed ratio. Therefore, with $C := \binom{K}{t}\binom{N-K}{t-1}\binom{N-K-t+1}{2}/\big[\binom{N}{k}\binom{N-k}{2}\big] \ge 0$,

$$\mathbb{P}(E_2) - \mathbb{P}(F_0) = C\big[(K-t) - (N-K-t)\big] = C\,(2K - N) \;\ge\; 0, \tag{5}$$

because $K > N/2$ in Case R. The excluded support case was already proved directly; if $t > K$, $\mathbb{P}(x_k = t-1) = 0$ and both atoms vanish. By (4) this proves (3) and, by the reflection identity, $U_K(k+2) \le U_K(k)$ as well. Together with the even-N center case, this completes the proof of monotonicity along odd k. $\square$

Remark 4 (Verification at $N = 32$). Independently of the proof, the revision's exact-rational CPU audit enumerates all admissible (N, K, k) for $2 \le N \le 64$, including the boundary $(N, K, k) = (32, 17, 29)$, checks (4), the support split, and the monotonicity inequality; every even-N center is $1/2$ on the odd grid, and a center point mass with $\alpha < 1/2$ invokes the explicit full-N fallback. The historical $N = 32$ artifact remains a separate frozen result check.

Remark 5 (Why only odd k enters the certificate). Requiring k odd removes tie-breaking from the certificate family (every prefix has a strict majority side). Its binary search is only over a nonempty feasible odd ladder; FULL-N is a separate zero-flip fallback. Nothing else in the paper uses monotonicity at even k.

A.2 Oracle terminal-stop bookkeeping (Theorem 1)

For the oracle theorem, the stopping rule is $\tau = \min\{k \in \{3, \dots, N\} : c_{H_\star}(x_k, k) \le \alpha\}$. At a reachable terminal state, all N binary outcomes have been observed, hence $x_N = K$. We define the endpoint certificate to be zero; whenever the posterior is supported, this also follows because the hypergeometric likelihood at (x_N, N) has support only on $K = x_N$:

$$c_{H_\star}(x_N, N) = 0.$$

Thus $k = N$ satisfies the same threshold rule rather than being an exceptional terminal state. The tower argument applies to this bounded stopping time, and

$$\mathbb{P}_{H_\star}(\text{flip}) = \mathbb{E}_{H_\star}\big[c_{H_\star}(x_\tau, \tau)\big] = \mathbb{E}_{H_\star}\big[c_{H_\star}\, \mathbf{1}\{\tau < N\}\big] + \mathbb{E}_{H_\star}\big[c_{H_\star}\, \mathbf{1}\{\tau = N\}\big].$$

The first term is at most $\alpha\, \mathbb{P}(\tau < N)$, and the second term is zero. Therefore the integrand is bounded pointwise by α and $\mathbb{P}(\text{flip}) \le \alpha$. For each fixed true count K, a path reaching N also has $x_N = K$, so its terminal contribution to the DP flip probability $g(K; \alpha, \widehat{H}_m)$ is exactly zero. This endpoint fact also holds for the fitted DP, but it does not identify a fitted earlier-prefix score with the oracle conditional probability.

Remark 6 (The terminal-stop mass is small in practice). On the TEST split, the exact fraction of paths reaching $k = N$ under BAYES-H is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$ (the fitted score stops early on the reported bimodal mass). Their contribution to replay flip is zero; their contribution to the expected stopping count is included exactly (Appendix B).

A.3 Linearity of the population flip rate

For the population statement, FIT and CAL are assumed independent i.i.d. task samples from the same count-exchangeable replay population. Conditional on FIT, every prior, fitted-score table, and candidate rule is frozen before CAL is inspected. For such a fixed rule and fixed prior $\widehat{H}_{\text{FIT}}$ used *inside* the fitted score, the per-task flip probability $g(K)$ depends only on the true count K (via the hypergeometric law of prefixes) and on $\widehat{H}_{\text{FIT}}$ (through the posterior threshold). The CAL values $g(K_i)$ are consequently i.i.d. bounded observations conditional on FIT, and the population replay flip rate is

$$\mathbb{P}_{\text{replay}, H_\star}(\text{flip}) = \sum_{K=0}^{N} H_\star[K]\, g(K),$$

a *linear* functional of the true mixture $H_\star$. Conditional on FIT, empirical Bernstein on the CAL mean bounds the population quantity for a frozen rule, with each per-task quantity $g \in [0, 1]$. Bonferroni over the pre-declared family of $J = 64$ rules permits CAL-side selection because simultaneous UCBs cover all J frozen linear functionals. The fixed benchmark TEST split is not used in this argument and remains descriptive. This population result is distinct from the oracle conditional identity in Theorem 1.

B Exact dynamic program for the adaptive rule

For a fixed true count K and frozen fitted prior $\widehat{H} = \widehat{H}_m$, we compute the exact per-task replay-flip probability $g(K; \alpha, \widehat{H})$ and expected stopping time $\mathbb{E}[\widehat{\tau} \mid K]$ of the resulting BAYES-H fitted-score rule under the replay model, by dynamic programming over prefix states.

State space and transitions. A state is a prefix (k, x) with $0 \le x \le k \le N$, $k \ge 0$. Under count K, the next draw is a uniform element of the remainder, so the transition kernel is

$$\mathbb{P}(x_{k+1} = x + 1 \mid k, x, K) = \frac{K - x}{N - k}, \qquad \mathbb{P}(x_{k+1} = x \mid k, x, K) = \frac{N - K - (k - x)}{N - k},$$

and the initial state has $\mathbb{P}(x_0 = 0) = 1$. The state graph is a DAG ($O(N^2)$ states, $O(N^2)$ edges), so a single backward pass is linear in the number of states.

Fitted-score table. The plug-in posterior $\mathbb{P}_{\widehat{H}}(K = j \mid k, x) \propto \widehat{H}[j]\,\mathrm{HG}(N, j, k, x)$ and fitted score $s_{\widehat{H}}(x, k)$ are precomputed once for all (k, x) on the $O(N^2)$ grid (with terminal convention $s_{\widehat{H}}(x, N) := 0$). At a posterior-supported endpoint, the likelihood supports only $j = x$, so the terminal score is zero; it also defines a zero-prior-mass table row, where observing all outcomes fixes the full-N binary pass-count decision. A state is a *stop state* iff $k \ge 3$ and $s_{\widehat{H}}(x, k) \le \alpha$; therefore $k = N$ satisfies the same stopping test rather than a separate truncation rule. This fitted score is not called the oracle conditional probability unless $\widehat{H} = H_\star$.

Recursions. For a fixed K, define $V(k, x) = \mathbb{P}(\mathrm{MV}(\tau) \ne \mathrm{MV}(N) \mid k, x, K)$ and $T(k, x) = \mathbb{E}[\tau - k \mid k, x, K]$. Then

$$V(k, x) = \begin{cases} \mathbf{1}\{\mathrm{side}(x, k) \ne \mathrm{side}(K, N)\}, & (k, x) \text{ stop state}, \\ \dfrac{K - x}{N - k}\, V(k + 1, x + 1) + \dfrac{N - K - k + x}{N - k}\, V(k + 1, x), & \text{otherwise}, \end{cases}$$

and the same recursion with $T(\cdot) = 0$ at stop states and $T(k, x) = 1 +$ (continuation mean) otherwise. The requested quantities are $g(K) = V(0, 0)$ and $\mathbb{E}[\tau \mid K] = T(0, 0)$. The per-task exact saving is $1 - \mathbb{E}[\tau \mid K]/N$; population quantities follow by linearity, $\sum_K H[K] g(K)$, for any mixture H. The same forward pass yields the exact terminal-stop occupancy $\mathbb{P}(\tau = N \mid K)$. At that boundary the reachable state has $x = K$, so the stop-state value in the recursion is zero for replay flip; on the TEST split terminal occupancy is 0.59% at $\alpha = 0.05$ and 1.10% at $\alpha = 0.02$.

Complexity. For each K: $O(N^2)$ states, $O(1)$ work per state after the fitted-score table is frozen, so all 33 values of K cost $O(N^3) \approx 3.6 \times 10^4$ flops at $N = 32$ — negligible. This is why every TEST number in the main table is an *exact replay quantity*, not a Monte Carlo estimate.

Cross-check against simulation. The DP is validated against an independent replay Monte Carlo implementation (fresh random prefixes) on the same frozen prior: the maximum absolute deviation over the checked cells is 0.043, within the pre-set tolerance $3\sigma_{\mathrm{MC}} = 0.075$ (the check includes cells at $\alpha = 0.01$, whose realized flips are small and inflate the pooled MC standard error); PASS. See `fit_cal_test_r469.py` and the self-check block of `fit_cal_test_r469_result.json`.

C Feature comparison and appendix roadmap

This appendix separates proof (Appendix A), exact DP bookkeeping (Appendix B), synthetic diagnostics, and fixed-rule sensitivity. None expands the main count-exchangeable replay claim to online correctness or operational cost.

D Full drift tables

Tables 3 and 4 report the complete drift stress grid behind Section 7 (`drift_stress_r469_result.json`; 2000 TEST problems, 500 orders per cell). These are synthetic exploratory diagnostics, not confirmatory or natural-order results. In particular, at E3, $\alpha = 0.05$, $\delta = 0.15$, BAYES-H is invalid (5.2%) while FIXED-EB is valid (4.7%); the tables support no global robustness ranking.

Work	Endpoint	Order/model	Guarantee / selection
ASC (Aggarwal et al., 2023); ESC (Li et al., 2024)	adaptive self-consistency	implementation-specific sequential sampling	no claim here about their universal full-vote certificate
MARS (Chen et al., 2026)	parallel trace vote movement	parallel traces/checkpoints	conservative future-movement bound; distinct from this replay posterior
Conformal (Vovk et al., 2005); CS (Howard et al., 2021)	coverage / time-uniform mean bounds	exchangeable / i.i.d.-sequential	established marginal or anytime tools, not this conditional flip object
Waudby-Smith–Ramdas (Waudby-Smith & Ramdas, 2020)	full-N majority via a WoR-CS stopper	finite population, uniform random prefix	direct same-endpoint comparator; not evaluated here, so no efficiency separation is claimed
This work	full-N binary pass-count replay disagreement	count-exchangeable random prefix	oracle conditional score under $H_\star$; fitted-score FIT/CAL screening; descriptive TEST replay readout

Table 2: Source-verified feature comparison. Relationships are deliberately endpoint- and model-specific; this table makes no priority or ordered-online claim.

Table 3: Synthetic exploratory/nonconfirmatory drift stress table at $\alpha = 0.05$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold**: realized flip exceeds α.

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	2.5 (6.1)	3.2 (17.0)	1.6 (27.0)	**5.8 (4.4)**
	0.05	4.8 (6.1)	4.9 (17.0)	4.1 (27.0)	**7.1 (4.4)**
	0.1	**7.6 (6.2)**	**8.2 (17.0)**	**7.6 (27.0)**	**9.6 (4.3)**
	0.15	**11.4 (6.0)**	**12.1 (17.0)**	**12.4 (27.0)**	**12.2 (4.2)**
	0.2	**13.0 (5.5)**	**13.7 (17.0)**	**13.6 (27.0)**	**13.5 (4.1)**
E2 (linear trend)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.05	4.8 (6.2)	4.7 (17.0)	3.7 (27.0)	**7.0 (4.4)**
	0.1	**5.2 (6.3)**	4.9 (17.0)	3.7 (27.0)	**7.6 (4.5)**
	0.15	**5.9 (6.4)**	**5.3 (17.0)**	3.7 (27.0)	**8.8 (4.6)**
	0.2	**7.0 (6.5)**	**6.0 (17.0)**	3.7 (27.0)	**10.4 (4.6)**
E3 (adversarial block)	0.0	4.7 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.05	4.8 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.1	4.9 (6.1)	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.15	**5.2 (6.1)**	4.7 (17.0)	3.8 (27.0)	**6.8 (4.4)**
	0.2	**5.5 (6.1)**	4.7 (17.0)	4.2 (27.0)	**6.8 (4.4)**

E THE CRITICAL-RADIUS CONSERVATION CURVE

Figure 2 is a deterministic visualization of the stored fitted-rule sensitivity curves. Within a constant fitted-rule interval, g and hence $V(R)$ are fixed; rule changes create the displayed jumps. The listed OpenR1 thresholds are algebraic properties of its frozen, three-atom fitted rule. The plotted carrier comparisons and alignment statistics are descriptive diagnostics of these stored curves, not a general cross-carrier relationship, a target-population statement, or a prescription for choosing α. The nested-FIT-size and rule/prior-channel calculations supplied with the snapshot are retained only as fitted-rule sensitivity diagnostics. They do not estimate a population shift, establish a FIT-size requirement, validate a margin repair, or yield a policy-selection recommendation. Their numerical details therefore do not enter the paper's current evidence claims. *(g) Oracle K-wise profile-capping sensitivity (withdrawn as a policy claim).* The stored fitted-score-ordered subset calculation selects a separate set $S_K(c)$ after conditioning on each final count K and then evaluates the coordinate $g_{S_K}(K)$. It is therefore an *oracle K-wise profile-capping sensitivity calculation*, not the flip profile

Table 4: Synthetic exploratory/nonconfirmatory drift stress table at $\alpha = 0.1$ (2000 TEST problems, 500 orders each; flip in %, mean budget k in parentheses). **Bold**: realized flip exceeds α.

Mechanism	δ	BAYES-H	FIXED-EB	FIXED-HOEF	WINDOW3
E1 (front-load)	0.0	3.7 (5.1)	7.9 (5.0)	6.4 (7.0)	5.8 (4.4)
	0.05	5.6 (5.1)	9.1 (5.0)	7.8 (7.0)	7.1 (4.4)
	0.1	8.4 (5.2)	**11.0 (5.0)**	**10.1 (7.0)**	9.6 (4.3)
	0.15	**11.9 (5.0)**	**12.7 (5.0)**	**12.3 (7.0)**	**12.2 (4.2)**
	0.2	**13.3 (4.7)**	**13.7 (5.0)**	**13.5 (7.0)**	**13.5 (4.1)**
E2 (linear trend)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.5 (5.2)	8.9 (5.0)	7.5 (7.0)	7.0 (4.4)
	0.1	6.0 (5.3)	10.0 (5.0)	8.3 (7.0)	7.6 (4.5)
	0.15	6.9 (5.3)	**11.6 (5.0)**	9.7 (7.0)	8.8 (4.6)
	0.2	8.3 (5.4)	**13.7 (5.0)**	**11.6 (7.0)**	**10.4 (4.6)**
E3 (adversarial block)	0.0	5.4 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.05	5.4 (5.1)	8.6 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.1	5.5 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)
	0.15	5.6 (5.1)	8.7 (5.0)	7.4 (7.0)	6.8 (4.4)
	0.2	5.7 (5.1)	8.7 (5.0)	7.3 (7.0)	6.8 (4.4)

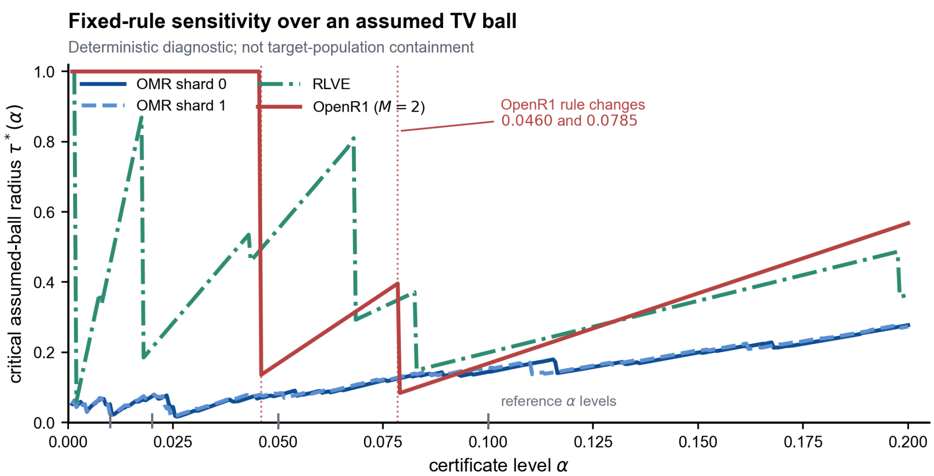

Figure 2: Deterministic fixed-rule sensitivity radius $\tau^*(\alpha)$ over an assumed fitted-mixture TV ball; this is not a confidence set for, or a containment claim about, an unknown target population. Proposition 1 is evaluated on the continuum $\alpha \in [10^{-3}, 0.2]$ (step 5×10^{-4}), computed exactly per level from the frozen g-profiles of the three carriers (OMR both FIT shards; OpenR1 exact in closed form). Grey tick marks: the four reference levels; red dotted lines: the OpenR1 fitted-score thresholds $0.0785/0.0460$ where the frozen rule changes. Artifact: `tv_conservation_r484_result.json`.

of one rule. The final count K is unobserved at every nonterminal prefix, and the artifacts provide neither a common state table $S(x, k)$ nor a proof that their K-indexed memberships agree on shared reachable prefix states. Consequently, this calculation is not a stopping policy, not prefix-observable, and not an instance of Theorem 1. It cannot support executable or deployable repair, policy-level count/cost claims, or claims about extra votes, universal budgets, 70-of-72 cells, domination, edge laws, crossing sets, or full-simplex robustness. Those prior numerical interpretations are withdrawn; no new calculation is reported here. Establishing any policy-level repair would require one K-independent table $S(x, k)$ built only from observable prefix information, a prefix-measurability check, and recomputation of every profile and capacity-aware TV LP from that same table.

F CROSS-CARRIER REPLAY EVIDENCE MATRIX

Table 5 collects the supplied in-distribution replay readouts: the four named model–carrier settings, named rules, and one descriptive α=0.01 CAL-side row for OMR (where the reported TEST family ends at α=0.02). Every flip is replay agreement with a corresponding full vote and every saving is replay count reduction; it is not gold correctness or operational cost. "CAL EB UCB" means a FIT-frozen empirical-Bernstein UCB with the stated Bonferroni allocation, not an oracle conditional certificate; TEST is a single descriptive readout. Artifacts: `fit_cal_test_r469`, `shard1_robust_r471` (R1/R2), `openr1_m2_pilot_r473`, `rlve_n8_r474`.

Table 5: Replay evidence matrix. "CAL EB UCB" is FIT-frozen and Bonferroni-adjusted over the pre-declared family; it is not a TEST guarantee or an oracle conditional certificate. "null" = no CAL-screened replay budget at that level; "—" = not applicable / not reported. FULL rows are the zero-count-reduction reference. Seeds 20260815 throughout.

Carrier	rule	α	CAL EB UCB	descriptive TEST flip	descriptive TEST mean k	descriptive TEST reduction
OMR s0 (N=32)	FIXED-HOEF	.10	0.0970	0.0645	7.0	78.1%
	FIXED-EB	.10	0.0958	0.0794	5.0	84.4%
	BAYES-H	.10	0.0495	0.0370	5.1	84.0%
	FIXED-HOEF	.05	0.0471	0.0156	27.0	15.6%
	FIXED-EB	.05	0.0461	0.0324	17.0	46.9%
	BAYES-H	.05	0.0350	0.0249	6.1	80.9%
	FIXED-HOEF	.02	null	—	—	—
	FIXED-EB	.02	0.0171	0.0083	31.0	3.1%
	BAYES-H	.02	0.0163	0.0091	8.5	73.5%
	BAYES-H	.01	0.0119	*CAL-only; no TEST readout*		
	FULL-32	—	—	0	32.0	0
OMR s1 (N=32)	FIXED-HOEF	.05	0.0488	0.0171	25.0	21.9%
	FIXED-EB	.05	0.0486	0.0357	15.0	53.1%
	BAYES-H	.05	0.0340	0.0246	6.2	80.6%
	BAYES-H†	.05	0.0341	0.0247	6.2	80.6%
	BAYES-H	.02	0.0158	0.0088	8.6	73.1%
	BAYES-H†	.02	0.0159	0.0089	8.6	73.2%
OpenR1 (M=2)	FIXED-1	—	0.0707	0.0592	1.0	50.0%
	BAYES-H	.10	0.0707	0.0592	1.0	50.0%
	BAYES-H	.05	0.0383	0.0294	1.4	32.1%
	BAYES-H	.02	0.0047	0	2.0	0
RLVE (N=8)	FIXED-HOEF	.10	0.0942	0.0613	5.0	37.5%
	FIXED-EB	.10	0.0783	0.0613	5.0	37.5%
	BAYES-H	.10	0.0469	0.0314	3.4	57.6%
	FIXED-EB/HOEF	.05–.01	null	—	—	—
	BAYES-H	.05	0.0184	0.0103	3.9	50.9%
	BAYES-H	.02	0.0121	0.0046	4.1	48.2%
	BAYES-H	.01	0.0081	0.0019	4.4	45.4%

†Cross-shard prior transfer (prior fit on shard 0 FIT, TV=0.042 to the shard 1 prior); all numbers move $\leq$0.4 points against within-shard. All CAL EB UCB cells use a FIT-frozen rule and the applicable pre-declared Bonferroni family. WINDOW3 has no CAL UCB and is omitted from the matrix (its reported replay flip is 0.0577; Table 1).

Scope of the matrix. The rows are a descriptive collation of supplied count-replay artifacts. They do not define or exhaust a feasible policy class, so they do not establish a broader ordering or a general relationship among carrier structure, fitted scores, and replay count reduction. The synthetic drift/margin rows elsewhere in the paper remain synthetic, exploratory, and nonconfirmatory; the E3 $\alpha = 0.05$, $\delta = 0.15$ cell where FIXED-EB is valid and BAYES-H is invalid is retained as a counterexample to a global ranking. Fitted-mixture TV quantities remain assumed-ball sensitivity diagnostics, not transfer guarantees.